VINAY SAMUEL

vsamuel@umd.edu 713-705-6174 Google Scholar LinkedIn GitHub

EDUCATION

University of Maryland, College Park *2025–2026*
Master of Science in Computer Science GPA: 4.00

Carnegie Mellon University *2021–2025*
Bachelor of Science in Statistics and Machine Learning

WORK EXPERIENCE

Scale AI San Francisco, CA
Machine Learning Research Intern May 2026–Present

- Leading research on **self-improving deep research coding agents** for enterprise applications.
- Designing an **unsupervised skill acquisition framework** in which agents generate synthetic skills to guide environment exploration and then apply accumulated skills to incoming user tasks thereby enabling zero-shot generalization to novel enterprise coding problems
- Architecting a **continual skill refinement** mechanism that allows agents to acquire and update skills during live user interactions, creating a self-improving loop without requiring supervised labels

RESEARCH EXPERIENCE

University of Maryland CLIP Group College Park, MD
Research Engineer, Advised by Prof. Mohit Iyyer Jan 2026–May 2026

- Designed **REDIPO**, an offline **DPO** data construction pipeline to recover distributional diversity in post-trained LLMs while preserving alignment, achieving up to **134% improvement** in response diversity (NoveltyBench distinct-k) across 3 model families
- Engineered **marginal diversity scoring** via embedding-based cosine similarity and quality-bounded preference pair construction to isolate diversity signal from quality thereby preventing DPO from collapsing to quality-only optimization
- Fine-tuned **Qwen3-4B, OLMo-3-7B, and LLaMA-3.1-8B** with **LoRA** and DPO; improved **safety** (lowest HarmBench ASR on all models) while maintaining instruction-following quality (MTBench, IFEval)

Stanford NLP Group Palo Alto, CA
Research Scientist, Advised by Prof. Diyi Yang June 2024–Feb 2025

- Built **Co-Gym**, an interactive platform for **human-AI collaboration** across travel planning, literature reviews, and data analysis - achieving **86% win rate in travel planning** vs. autonomous baselines
- Developed **production-ready Related-Works generation system** enabling dynamic interaction between researchers and LLM agents for automated literature synthesis

CMU LTI Pittsburgh, PA
Lead ML Research Scientist, Advised by Prof. Daphne Ippolito May 2024–May 2025

- Engineered novel **control framework** for precise attribute control in LM generation (word count), achieving up to **58.54% improvement** in constraint satisfaction
- Scaled model training pipeline using **LoRA and full fine-tuning** of LLaMA-3-8B with **Accelerate, DDP, and Deepspeed** - reducing training time by **60%**
- Curated **50k instruction-tuning dataset** for benchmarking controllable generation for model evaluation

Princeton NLP Group Princeton, NJ
Lead ML Research Scientist, Advised by Prof. Karthik Narasimhan Mar 2024–Jan 2025

- Built the **PersonaGym evaluation platform** and PersonaScore metric for **LLM persona agent evaluation** - deployed by **30+ researchers** for model assessment
- Benchmarked **10 state-of-the-art LLMs** across 200 personas with 10K evaluation questions, identifying key performance gaps
- Optimized **prompt engineering pipeline**, reducing evaluation cost by **70%** while maintaining accuracy

SELECTED PUBLICATIONS

- **[LLM]** **Vinay Samuel**, Yapei Chang, Mohit Iyyer. *Recovering Diversity Without Losing Alignment: A DPO Recipe for Post-Trained LLMs* **Under Review** [Paper]
- **[Agent]** Yijia Shao, **Vinay Samuel**, Yucheng Jiang, John Yang, Diyi Yang. *Collaborative Gym: A Framework for Enabling and Evaluating Human-Agent Collaboration.* **ICLR 2026** [Paper]
- **[LLM]** **Vinay Samuel**, Harshita Diddee, Yiming Zhang, Daphne Ippolito. *CIE: Controlling Language Model Text Generations Using Continuous Signals.* **EMNLP 2025 (Main)** [Paper]
- **[Agent]** **Vinay Samuel**, Henry Peng Zou, Yue Zhou, ..., Ameet Deshpande, Karthik Narasimhan, Vishvak Murahari. *PersonaGym: Evaluating Persona Agents and LLMs.* **EMNLP 2025 (Findings)** [Paper]
- **[LLM]** **Vinay Samuel**, Yue Zhou, Henry Peng Zou. *Towards Data Contamination Detection for Modern Large Language Models: Limitations, Inconsistencies, and Oracle Challenges.* **COLING 2025** [Paper]
- **[Evaluation]** Yiming Zhang, Harshita Diddee, ..., **Vinay Samuel**, Barry Wang, Daphne Ippolito. *NoveltyBench: Evaluating Language Models for Human-like Diversity.* **COLM 2025** [Paper]
- **[Multimodal LLM]** Henry Peng Zou, **Vinay Samuel**, Yue Zhou, Weizhi Zhang, Liancheng Fang, Zihe Song, Philip S. Yu, Cornelia Caragea. *ImplicitAVE: An Open-Source Dataset and Multimodal LLMs Benchmark for Implicit Attribute Value Extraction.* **ACL 2024 (Findings)** [Paper]
- **[Data Generation]** **Vinay Samuel**, Houda Aynaou, ..., Aman Chadha. *Can LLMs Augment Low-Resource Reading Comprehension Datasets? Opportunities and Challenges.* **ACL 2024 (SRW)** [Paper]

SKILLS

- **Research**: LLMs, Multimodal Learning, NLP, Agents, Semi-Supervised Learning, Retrieval Augmented Generation (RAG), Reinforcement Learning, Question Answering, LM Controllability, Evaluation, Fine-tuning, Model Parallelization, Data Preprocessing, Data Analysis, GPU, CPU, Self-Improvement
- **Languages and Platforms**: Python, PyTorch, R, LaTeX, MATLAB, Colab, Git, Bash, SQL, C++, Triton
- **Soft Skills**: Self-Motivated, Team-Spirited, Leadership, Critical Thinking, Collaboration, Analytical, Problem Solving, Strong Communicator

COURSES

- **LLM & Agent**: Large Language Models, AI Problem Solving and Representation
- **Machine Learning**: Introduction to Machine Learning, Convex Optimization, Introduction to Deep Learning
- **Natural Language Processing**: Advanced Natural Language Processing
- **Systems**: Systems for Machine Learning